Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/16/2005
PI Name:	Talley, Douglas G Ph.D.		
Project Title:	Supercritical and Transcritical Shear Flows in Microgravity: Experiments and Direct Numerical Simulations		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Douglas.talley@edwards.af.mil	**Fax:**	FY 661-275-6245
PI Organization Type:	GOVERNMENT	**Phone:**	661-275-6174
Organization Name:	Air Force Research Laboratory		
PI Address 1:	AFRL/PRSA		
PI Address 2:	10 East Saturn Boulevard		
PI Web Page:			
City:	Edwards AFB	**State:**	CA
Zip Code:	93524-7661	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	04/30/2002	**End Date:**	04/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
Contact Email:	walter.m.duval@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Bellan, Josette Ph.D. (Jet Propulsion Laboratory) Chehroudi, Bruce Ph.D. (ERC Inc.)		
Grant/Contract No.:	C3211OT		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this research is to develop and experimentally validate a near-critical transcritical and supercritical fluid shear flow model independent of turbulence. We define a supercritical shear flow to be one in which all of the fluid particles remain above their critical temperature and pressure. We define a transcritical shear flow to be one in which at least some of the fluid particles undergo a transition between a subcritical and a supercritical temperature, between a subcritical and a supercritical pressure, or both. The reason it is necessary to validate the fluid model independent of turbulence is that turbulence introduces a large number of additional mechanisms the understanding of which is embryonic at best for near-critical transcritical and supercritical flows. Validating the fluid model without turbulence uncertainties therefore requires laminar flows. However, laminar flows that are not influenced by gravity are difficult to produce in normal gravity due to the large density gradients involved. Therefore, microgravity experiments are necessary. Experiments consisting of visualizing perturbed and un-perturbed laminar jets in microgravity will be conducted in the ground-based facilities at NASA GRC. Direct numerical simulations of the experiments will be performed at the Jet Propulsion Laboratory. Perform high speed visualizations of perturbed and unperturbed laminar, round, cryogenic jets of liquid nitrogen injected into room temperature gaseous nitrogen and gaseous helium in microgravity at various supercritical pressures. Ensure the experiments are performed such that all initial and boundary conditions are known. Validate a fluid shear model by performing direct numerical simulations of the experiments. Injection processes involving transcritical and supercritical shear flows occur in a number of applications where high pressures contribute to performance benefits. Examples include liquid rocket engines, gas turbines, internal combustion engines, and chemical processing applications. Such shear flows are characterized by very steep density gradients approaching gas/liquid systems, but without surface discontinuities, and require sophisticated treatments of transport properties and the equation of state. A scientific understanding of the physics of these flows does not exist. This will be the first time model containing all the necessary physics will have been directly compared to experimental data.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	An experimental investigation was conducted on a coaxial jet similar to those used in cryogenic liquid rockets, spanning sub- to supercritical conditions, with the intent of gaining a better understanding of an aspect of combustion instability that pertains to interactions of an externally-imposed acoustic field with the jet in normal gravity. Past research works on this subject have shown both the relevance and importance of geometrical changes in injector exit-area and its nearby physical and fluid mechanical processes. Therefore, special attention is paid in collecting spatially-resolved mean temperatures at the exit of this injector. High frame rate digital cameras provided information on the behavior of this jet under variety of conditions. Mean and rms values of the “dark core” length fluctuations were measured from the acquired images via a computer-automated method. There appears to be a good correlation between this length and the outer-to-inner jet momentum ratio, but the form of this dependence was found to be different at subcritical than the rest of the conditions. The rms values of the length fluctuations suggested a possible explanation for the observed improvement in instability limit at increasingly higher outer-to-inner jet velocity ratios. According to the plan formulated at the end of last year for the numerical portion of the work, using low-Mach-number asymptotics, we formally derived this year modified governing equations (“low-Mach-number formulation”) that can be solved at larger time-steps than original equations. The scheme can now solve the original governing equations (“compressible formulation”), by integrating different variables, in a numerically stable manner at low Mach numbers, and are using analyses of both the simulation results and the governing equations in order to determine a set of modified equations that retains the physics of interest while allowing larger time-steps. The assumptions used to derive the low-Mach-number formulation from the compressible formulation were verified to be reasonable based on simulations using the compressible formulation; the low-Mach-number formulation did not yield physically consistent results with the compressible formulation. The lack of agreement was attributed to failure of the low-Mach-number formulation to properly represent the pressure field. Therefore the low-Mach-number approach is not feasible for this problem, and it is necessary to proceed with the compressible formulation. For all low-Mach-simulations it was observed that the results at the lowest reasonable Mach number for the compressible formulation (0.05) were very similar to those at the target experimental Mach number, except for the pressure field which scales as Mach number squared, the indications are that the Mach=0.05 results will be generally applicable to the lower Mach numbers. These indications will be appropriately verified once comparisons between experimental results and simulations are performed, at which point the issue of computational Mach number may be revisited. To enable objective comparisons between simulations we have obtained a quantitative criterion to distinguish vortical structures in images, similar to those that would be provided by experiments, and showed that unlike other criteria proposed for incompressible low-pressure flows, the selected criterion can identify the coherent structures expected from high-pressure flows.
Bibliography Type:	Description: (Last Updated: 05/16/2006)
Awards	Okong'o, N., and Bellan, J. "NASA Certificate of Recognition" Jan-2005
Dissertations and Theses	Davis, D. "Acoustically Driven Shear-Coaxial Jet Studies under Sub-, Near-, and Supercritical Conditions" Ph.D. comprehensive exam Apr-2005
NASA Technical Documents	Okong'o, N., and Bellan, J. "LES of Supercritical Temporal Mixing Layers: A Priori Analysis" Seleced for a NASA Tech Brief Apr-2005
Papers from Meeting Proceedings	Dustin Davis, Bruce Chehroudi, and Ira Sorensen "Measurements in an Acoustically Driven Coaxial Jet Under Supercritical Condition" 43rd AIAA Aerospace Sciences Meeting and Exhibit, Reno, January 2005, Paper AIAA-2005-0736 Jan-2005
Papers from Meeting Proceedings	Okong'o, N., and Bellan, J. "LES of Supercritical Temporal Mixing Layers: A Priori Analysis" 43rd AIAA Aerospace Sciences Meeting and Exhibit, Reno, January 2005, Paper AIAA-2005-0155 Jan-2005